

BLORAT

BLAT-like ORF Record Alignment Tool

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Abstract

BLAT (BLAST-Like Alignment Tool) is a bioinformatics software tool for comparing nucleotide or protein sequences against large genomic databases. Developed by Jim Kent from UCSC, it is designed as a speed-optimized version of BLAT for his use cases, quickly finding sequences of 95% and greater similarity of length 40 bases or more. This makes it highly efficient for mapping DNA sequences to genomes or finding protein homologs.

Integrating data from BLAT with findings from an Open Reading Frame (ORF) finder can be useful in genomic studies. ORF finders identify regions within a DNA or RNA sequence that could encode proteins, indicating possible gene locations. However, determining the function or relevance of these ORFs and ensuring they map accurately to known genomic locations requires further validation. This is where BLAT comes in; it can align these identified ORFs against a genomic database, confirming their existence in the genome and helping annotate them based on similarity to known genes. This complementary approach enhances the identification and annotation of genes, improving our understanding of genomic structures and functions. Extending the lab 5 ORF finder to incorporate BLAT searches would allow for this extension, creating a final program lovingly named BLORAT. BLAT-like ORF Record Alignment Tool.

Introduction of the Problem

In conversations with Hiram M Clawson, Sr. Software Developer at the Genomics Institute of UCSC, there are significant numbers of genetic counselors who use the genome browser and its' tools including BLAT in their work to support patient healthcare outcomes. In fact, this usage became quite apparent during a recent outage that resulted in a strong outpouring of emails to his department about the impacts of the outage from genetic counselors.

Description of the Solution

BLORAT aims to provide seamless integration between ORF identification and genomic alignment, enhancing the usability of local BLAT installations for genetic counselors and researchers. The program will automate the process of taking ORFs identified by an integrated

ORF finder, running them through BLAT for genomic alignment, and producing user-friendly output that includes annotated genomic locations and potential functional information. The steps involved in this solution include:

1. Identifying potential ORFs within a given DNA sequence using an ORF finder.
2. Using BLAT to align these ORFs to a reference genome.
3. Parsing and annotating BLAT's output to integrate genomic location data and potential gene function information.

The output can be formatted into a CSV file or similar for further analysis. This approach will simplify data processing for genetic counselors, providing them with immediate, actionable insights from their sequencing data.

Discussion & Evaluation

The effectiveness of BLORAT will be evaluated through a few metrics:

1. Accuracy: Comparing BLAT alignment results with known genomic locations to assess precision. In this case, the alignments match 1:1
2. Speed: Measuring the time taken for ORF identification and alignment in large datasets, given the program works directly on local hardware it is much faster at scale, but it does still take time. Perhaps adding user feedback like a loading bar or progress bar if we could access the underlying logs of BLAT.
3. Usability: Surveying end-users, such as genetic counselors, for feedback on ease of use, interface functionality, and output comprehensibility. Additionally, for a stronger understanding of what specific formats and filters are effective and useful in their day-to-day work.

Tests done during development indicate that BLORAT can efficiently process sequencing datasets, providing quick and accurate alignments and annotations. However, further testing and refinement are required to ensure it works effectively with edge cases and improve extensibility through a real-world product testing/feedback stage as, at this stage, the program is more or less an MVP.

Conclusion

In conclusion, BLORAT successfully integrates ORF identification with BLAT alignment, streamlining the process of mapping and annotating potential genes. This tool enhances the efficiency and usability of BLAT for local applications, which can be useful for genetic counselors and in other clinical settings. Future work will focus on improving the program's performance and integrating it with other tools to provide even more comprehensive genomic analysis. This project has showcased the potential of combining different bioinformatics tools to achieve better results and support genetic clinical practice more effectively.

References can be seen dispersed within the comments of the code as relevant.